Statistics 6611

Theory and Applications of Linear Models

Summary Notes

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1 Matrix Preliminaries: Generalized Inverses

Theorem 1.1. *For any matrix A of rank $r \geq 1$, there exist nonsingular matrices P and Q such that*

$$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix},$$

where D is a nonsingular $r \times r$ diagonal matrix.

The matrix

$$G = Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P$$

satisfies $AGA = A$, where $F, H,$ and B are any matrices of appropriate dimensions.

Definition. A generalized inverse of a matrix A is any matrix G that satisfies $AGA = A$.

Subsequently, for any matrix A, we use A^- to denote an arbitrary generalized inverse of A.

Definition. The equations $A\underline{x} = \underline{y}$ are said to be consistent if they have at least one solution.

Theorem 1.2. *For any matrix A, a matrix G is a generalized inverse of A if and only if $\underline{x} = G\underline{y}$ is a solution to the equations $A\underline{x} = \underline{y}$ for every vector $\underline{y}$ for which the equations are consistent.*

Remark. Theorem 1.2 can be extended to the matrix equation $AX = Y$. That is, for any matrix A, a matrix G is a generalized inverse of A if and only if $X = GY$ is a solution to the matrix equation $AX = Y$ for every matrix Y for which the equation is consistent.

Theorem 1.3. *All solutions to a consistent matrix equation $AX = Y$ are generated by putting*

$$X = A^-Y + (I - A^-A)Z$$

and letting Z range over the space of all possible matrices (of appropriate dimensions), where I denotes the identity matrix (of appropriate dimensions).

Lemma 1.1. *A matrix G is a generalized inverse of a matrix A if and only if G' is a generalized inverse of A', where A' is the transpose of A.*

Corollary. If A is a symmetric matrix, then $(A^-)'$ is a generalized inverse of A. In particular, for any matrix X, $[(X'X)^-]'$ is a generalized inverse of $X'X$.

Lemma 1.2. *For any matrix X, $\mathrm{tr}(X'X) = 0$ implies $X = 0$.*

Lemma 1.3. *For any matrix X and for any matrices P and Q of appropriate dimensions, we have (i) $PX'X = QX'X$ implies $PX' = QX'$ and (ii) $X'XP = X'XQ$ implies $XP = XQ$.*

Lemma 1.4. *For any matrix X, $X'X(X'X)^{-}X' = X'$, that is, $X(X'X)^{-}$ is a generalized inverse of X'.*

Corollary 1. (i) $X'X[(X'X)^{-}]'X' = X'$, (ii) $X(X'X)^{-}X'X = X$, and (iii) $X[(X'X)^{-}]'X'X = X$.

Corollary 2. Corresponding to any matrix X, there exists a matrix B satisfying $X'XB = X'$.

Lemma 1.5. *Let X be any matrix and let B_1 and B_2 be any two matrices that satisfy $X'XB = X'$. Then, $XB_1 = XB_2$.*

Corollary 1. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, $XBX = B'X'X = X$.

Corollary 2. Let X be any matrix and let G_1 and G_2 be any two generalized inverses of $X'X$. Then, $XG_1X' = XG_2X'$.

Corollary 3. For any matrix X, $X(X'X)^{-}X' = X[(X'X)^{-}]'X'$, that is, $X(X'X)^{-}X'$ is symmetric.

Corollary 4. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, the matrix XB is symmetric.

Let $P_X = XB$ where B is a solution to $X'XB = X'$. Then, P_X has the following properties.

(1) P_X is invariant to the choice of solution B (by Lemma 1.5).
(2) P_X is symmetric (by Corollary 4 to Lemma 1.5).
(3) $P_X = X(X'X)^{-}X'$ (by Lemma 1.4).
(4) $P_XX = X$ (by Corollary 1 to Lemma 1.5).
(5) P_X is idempotent, that is, $P_X^2 = P_X$ [by (3) and (4)].

P_X is said to be an orthogonal projection matrix.

Definition. The column space of an $m \times n$ matrix A consists of all $m \times 1$ vectors spanned by the columns of A, that is, a vector $\underline{c}$ belongs to the column space of A if and only if there exists a vector $\underline{b}$ for which $\underline{c} = A\underline{b}$. Similarly, the row space of A consists of all $1 \times n$ vectors spanned by the rows of A.

Notation. We use $\mathcal{C}(A)$ to denote the column space of A and use $\mathcal{R}(A)$ to denote the row space of A.

Lemma 1.6. *(i) $\mathcal{C}(B)$ is a subspace of $\mathcal{C}(A)$ if and only if $B = AC$ for some matrix C. (ii) $\mathcal{R}(B)$ is a subspace of $\mathcal{R}(A)$ if and only if $B = CA$ for some matrix C.*

Theorem 1.4. *Let* $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ *be a partitioned matrix.*

(i) If $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$, then the partitioned matrix

$$\begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix}$$

is a generalized inverse of M, where $Q = B - DA^-C$.

(ii) If $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(B)$ and $\mathcal{R}(C)$ is a subspace of $\mathcal{R}(B)$, then the partitioned matrix

$$\begin{bmatrix} P^- & -P^-CB^- \\ -B^-DP^- & B^- + B^-DP^-CB^- \end{bmatrix}$$

is a generalized inverse of M, where $P = A - CB^-D$.

Theorem 1.5. *Let $W = R + D$, where $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(R)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(R)$ (as would be the case, for example, if R were nonsingular). Take S, T, and U to be any three matrices such that*

$$D = STU.$$

Then, the matrix

$$R^- - R^-ST(T + TUR^-ST)^-TUR^-$$

is a generalized inverse of W.

2 Classical (Fixed-Effects) Estimation Theory

2.1 Linear Models

Let $\underline{y}$ represent an $n \times 1$ vector of (scalar-valued) data points. We suppose that

$$\underline{y} = X\underline{\beta} + \underline{\epsilon},$$

where X is a given $n \times p$ "design" matrix, $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters that is contained in some given parameter space, and $\underline{\epsilon}$ is an $n \times 1$ vector of unobservable "residuals" or "deviations."

It is convenient to introduce various special cases of this "linear model."

Model I (Least squares model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; parameter space $= \left\{\underline{\beta} \colon \underline{\beta} \in R^p\right\}$ (where R^p denotes the Euclidean p-space).

Model II (Gauss-Markov model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 I$; parameter space $= \left\{\underline{\beta}, \sigma^2 \colon \underline{\beta} \in R^p, \sigma^2 > 0\right\}$.

Model III (Aitken model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given positive definite or positive semidefinite matrix; parameter space $= \left\{\underline{\beta}, \sigma^2 \colon \underline{\beta} \in R^p, \sigma^2 > 0\right\}$.

Model IV (Mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon} = Z_1\underline{b}_1 + \cdots + Z_m\underline{b}_m + \underline{e}$, where Z_i is a given $n \times q_i$ "design" matrix, $\underline{b}_i$ is a $q_i \times 1$ random vector such that $E(\underline{b}_i) = \underline{0}$ and $\mathrm{Var}(\underline{b}_i) = \sigma_i^2 I$, and $\underline{e}$ is an $n \times 1$ random vector such that $E(\underline{e}) = \underline{0}$ and $\mathrm{Var}(\underline{e}) = \sigma^2 I$, and where $E(\underline{b}_i\underline{e}') = 0$ and $E(\underline{b}_i\underline{b}_j') = 0$ for $i = 1, \ldots, m$, $j = 1, \ldots, m$, and $i \neq j$; parameter space $= \left\{\underline{\beta}, \sigma_1^2, \ldots, \sigma_m^2, \sigma^2 \colon \underline{\beta} \in R^p, \sigma_1^2 \geq 0, \ldots, \sigma_m^2 \geq 0,\ \sigma^2 > 0\right\}$. Here, $\sigma_1^2, \ldots, \sigma_m^2$, and σ^2 are called variance components.

Model V (Random-effects model): a special case of the mixed-effects model where $X = \underline{1}$ (a column vector of 1's).

Model VI (General mixed-effects model, with a linear variance structure): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = \theta_1 H_1 + \cdots + \theta_m H_m$, where $H_1, \ldots, H_m$ are given symmetric matrices; parameter space $= \left\{\underline{\beta}, \underline{\theta} \colon \underline{\beta} \in R^p, \underline{\theta} \in \Omega\right\}$, where $\underline{\theta} = (\theta_1, \ldots, \theta_m)'$ and Ω is the set of $\underline{\theta}$-vectors for which $\mathrm{Var}(\underline{\epsilon})$ is positive definite or positive semidefinite, or Ω is a given subset of that set.

Model VII (General mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that$E(\underline{\epsilon}) = \underline{0}$ and $\mathrm{Var}(\underline{\epsilon}) = V$, where the elements of V are functions of some m-dimensional parameter vector

$\underline{\theta}$; parameter space $= \{\underline{\beta}, \underline{\theta} \colon \underline{\beta} \in R^p, \underline{\theta} \in \Omega\}$, where Ω is the set of $\underline{\theta}$-vectors for which V is positive definite or positive semidefinite, or Ω is a given subset of that set.

In all of Models I—VII, the space of possible values for $\underline{\beta}$ is the Euclidean p-space. Limiting $\underline{\beta}$ to some subset of R^p gives a constrained version of each model. The kinds of limitations on $\underline{\beta}$ that are most commonly imposed are those determined by linear equality constraints or linear inequality constraints, such as $A\underline{\beta} = \underline{b}$ or $A\underline{\beta} \geq \underline{b}$.

Notation. Let $p^* = \text{rank}(X)$.

When $p^* = p$, the linear model is customarily referred to as a full-rank linear model.

2.2 Estimability

Definition. An estimator $t(\underline{y})$ of a linear function $\underline{\lambda}'\underline{\beta}$ is said to be linear if $t(\underline{y}) \equiv c + \underline{a}'\underline{y}$ for some constant c and some vector $\underline{a}$ of constants.

Definition. Under Models II—VII, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] \equiv \underline{\lambda}'\underline{\beta}$.

Lemma 2.1. *Under Models II—VII, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.*

Definition. Under Models II—VII, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Otherwise, it is said to be nonestimable.

Theorem 2.1. *Under Models II—VII, a necessary and sufficient condition for a linear function $\underline{\lambda}'\underline{\beta}$ to be estimable is that its coefficient vector $\underline{\lambda}'$ be expressible in the form $\underline{\lambda}' = \underline{a}'X$, that is, that $\underline{\lambda}'$ belong to $\mathcal{R}(X)$, that is, that $\underline{\lambda}'\underline{\beta}$ be expressible as $\underline{a}'X\underline{\beta}$, that is, that*

$$\text{rank}(X) = \text{rank}\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix},$$

where $\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}$ is the matrix X augmented by the row vector $\underline{\lambda}'$.

Corollary 1. Under Models II—VII, the elements of the vector $X\underline{\beta}$ are estimable.

Corollary 2. Under Models II—VII, if $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_r'\underline{\beta}$ are estimable, then any linear combination of them is estimable.

Corollary 3. If $p^* = p$, then for every $p \times 1$ vector $\underline{\lambda}$, $\underline{\lambda}'\underline{\beta}$ is estimable.

Definition. Linear functions $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_r'\underline{\beta}$ are said to be linearly independent if $\underline{\lambda}_1, \ldots, \underline{\lambda}_r$ form a linearly independent set of vectors or, equivalently, if the matrix $\Lambda = (\underline{\lambda}_1, \ldots, \underline{\lambda}_r)$ has rank r.

Theorem 2.2. *Under Models II—VII, there exists a set of p^* linearly independent functions $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_{p^*}'\underline{\beta}$ that are estimable. No collection of estimable linear functions can contain more than p^* linearly independent functions.*

Lemma 2.2. *For any matrices A and B (of appropriate dimensions),* $\text{rank}(AB) \leq \text{rank}(A)$ *and* $\text{rank}(AB) \leq \text{rank}(B)$.

Lemma 2.3. *For any matrix A,* $\text{rank}(A'A) = \text{rank}(A)$.

2.3 Least Squares Estimation

Definition. Under Models II—VII, a (simple or ordinary) least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes the sum of squares

$$Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}).$$

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ is that $\widehat{\underline{\beta}}$ satisfy the normal equations

$$X'X\widehat{\underline{\beta}} = X'\underline{y}.$$

Theorem 2.3. *The normal equations are consistent.*

Theorem 2.4. *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ if and only if $\widehat{\underline{\beta}}$ is a solution to the normal equations.*

Theorem 2.5. *Under Models II—VII, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations.

2.4 Geometric Considerations

Definition. Two $t \times 1$ vectors $\underline{a}$ and $\underline{b}$ are said to be orthogonal if their inner product $\underline{a}'\underline{b}$ equals zero.

Definition. Let l_1 be the directed line segment from the origin $\underline{0}$ to a point $\underline{a} \neq \underline{0}$, and let l_2 be the directed line segment from $\underline{0}$ to a point $\underline{b} \neq \underline{0}$. The angle θ formed by l_1 and l_2 is defined by

$$\cos\theta = \frac{\underline{a}'\underline{b}}{\left[\underline{a}'\underline{a}\right]^{1/2}\left[\underline{b}'\underline{b}\right]^{1/2}}, \quad 0 \leq \theta \leq \pi.$$

When $\theta = \pi/2$, that is, when $\cos\theta = 0$, that is, when $\underline{a}$ and $\underline{b}$ are orthogonal, the directed line segments (or the lines defined by the directed line segments) are said to be perpendicular.

Notation. Let $\mathcal{C}^{\perp}(X)$ represent the null space of X', that is, the set of all column vectors that are orthogonal to every column of X [or, equivalently, to every vector in $\mathcal{C}(X)$]. Similarly, $\mathcal{R}^{\perp}(X)$ denotes the set of all row vectors that are orthogonal to every row of X.

Let $\widehat{\underline{y}} = X\widehat{\underline{\beta}}$ and $\widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}} = \underline{y} - X\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.

Theorem 2.6. *Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition satisfying $\underline{y} = \underline{y}_1 + \underline{y}_2$, where $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^{\perp}(X)$. Moreover, this decomposition is given by $\underline{y}_1 = \widehat{\underline{y}} = X\widehat{\underline{\beta}}$ and $\underline{y}_2 = \widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}} = \underline{y} - X\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Corollary. If W is any matrix with its column space identical to that of the matrix X [that is, $\mathcal{C}(W) = \mathcal{C}(X)$], then $W\widehat{\underline{\tau}} = X\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any solution to $X'X\underline{\beta} = X'\underline{y}$ and $\widehat{\underline{\tau}}$ is any solution to $W'W\underline{\tau} = W'\underline{y}$.

Note: $\widehat{\underline{\beta}} \equiv (X'X)^{-}X'\underline{y}$ solves $X'X\underline{\beta} = X'\underline{y}$, implying that $\widehat{\underline{y}} = X\widehat{\underline{\beta}} = P_X\underline{y}$ and $\widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}} = (I - P_X)\underline{y}$

2.5 The Conjugate Normal Equations

Theorem 2.7. *Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the conjugate normal equations*

$$X'X\underline{r} = \underline{\lambda}$$

are consistent, that is, if and only if there exists an $\underline{r}$ such that $X'X\underline{r} = \underline{\lambda}$.

Corollary. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X(X'X)^{-}\underline{\lambda} = \underline{\lambda}$.

Theorem 2.8. *Under Models II—VII, the least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is equal to $\underline{r}'X'\underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.*

2.6 Expectations/Variances/Covariances of Least Squares Estimators

For two random vectors $\underline{x} = (x_1, \ldots, x_m)'$ and $\underline{z} = (z_1, \ldots, z_n)'$, we write $\mathrm{cov}(\underline{x}, \underline{z})$ for the $m \times n$ matrix whose (i, j)th element is the covariance between x_i and z_j. The matrix $\mathrm{Var}(\underline{x}) = \mathrm{cov}(\underline{x}, \underline{x})$ is called the variance-covariance matrix, or simply the covariance matrix, of $\underline{x}$.

Lemma 2.4. *If $\underline{x}$ is a random vector and L and M are matrices of constants having the same number of rows as $\underline{x}$ has elements, then*

$$\mathrm{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M, \text{ where } \Sigma = \mathrm{Var}(\underline{x}).$$

Corollary. $\mathrm{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \mathrm{Var}(\underline{x})$.

Definition. For a matrix of constants Λ having p rows, the vector $\Lambda'\underline{\beta}$ is said to be estimable if each of its elements is estimable.

Suppose that $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ (where Λ_1 and Λ_2 are matrices of constants having p rows each) are both estimable. Let $\underline{\widehat{\beta}}$ represent any solution to the normal equations, and take R_1 and R_2 to be any two matrices satisfying

$$X'XR_1 = \Lambda_1 \quad \text{and} \quad X'XR_2 = \Lambda_2.$$

The least squares estimators of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, that is, the vectors whose elements are the ordinary least squares estimators of the corresponding elements of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, are given by

$$\Lambda_1'\underline{\widehat{\beta}} = R_1'X'\underline{y} \quad \text{and} \quad \Lambda_2'\underline{\widehat{\beta}} = R_2'X'\underline{y}.$$

Under Model VII,

$$\mathrm{cov}(\Lambda_1'\underline{\widehat{\beta}}, \Lambda_2'\underline{\widehat{\beta}}) = \Lambda_1'(X'X)^-X'VX(X'X)^-\Lambda_2,$$

and, in particular,

$$\mathrm{Var}(\Lambda_1'\underline{\widehat{\beta}}) = \Lambda_1'(X'X)^-X'VX(X'X)^-\Lambda_1.$$

Under Model II,

$$\mathrm{cov}(\Lambda_1'\underline{\widehat{\beta}}, \Lambda_2'\underline{\widehat{\beta}}) = \sigma^2\Lambda_1'(X'X)^-\Lambda_2,$$

and, in particular,

$$\mathrm{Var}(\Lambda_1'\underline{\widehat{\beta}}) = \sigma^2\Lambda_1'(X'X)^-\Lambda_1.$$

2.7 The Gauss-Markov Theorem

Theorem 2.9. (**The Gauss-Markov Theorem**) *Under Models II—VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\underline{\widehat{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\underline{\widehat{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under Model II, the variance of $\underline{\lambda}'\underline{\widehat{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, $\underline{\lambda}'\underline{\widehat{\beta}}$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.*

Definitions. If $\underline{x}$ is a column vector of r real variables and A is an $r \times r$ matrix of constants, then we refer to the function

$$q(\underline{x}) = \underline{x}'A\underline{x}$$

as a quadratic form. The quadratic form $q(\underline{x})$ is said to be: positive definite if $q(\underline{x}) > 0$ for all nonzero real vectors $\underline{x}$; nonnegative definite if $q(\underline{x}) \geq 0$ all real vectors $\underline{x}$; and positive semidefinite $q(\underline{x})$ is nonnegative definite but not positive definite (i.e., $q(\underline{x}) = 0$ holds for some nonzero $\underline{x}$).

A square matrix A is said to be positive definite, nonnegative definite, or positive semidefinite, if A is symmetric and the associated quadratic form $\underline{x}'A\underline{x}$ is positive definite, nonnegative definite, or positive semidefinite, respectively.

Theorem 2.10. *Under Model II, suppose that $\Lambda'\underline{\beta}$ is any k-dimensional estimable vector and that $\underline{c} + A'\underline{y}$ is any vector of linear unbiased estimators of the elements of $\Lambda'\underline{\beta}$. Then, the matrix $\mathrm{Var}(\underline{c} + A'\underline{y}) - \mathrm{Var}(\Lambda'\widehat{\underline{\beta}})$ is nonnegative definite, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Corollary 1. Under the condition of Theorem 2.10,

$$\mathrm{tr}[\mathrm{Var}(\underline{c} + A'\underline{y})] \geq \mathrm{tr}[\mathrm{Var}(\Lambda'\widehat{\underline{\beta}})].$$

Corollary 2. Under the condition of Theorem 2.10,

$$\det[\mathrm{Var}(\underline{c} + A'\underline{y})] \geq \det[\mathrm{Var}(\Lambda'\widehat{\underline{\beta}})].$$

Definition. The determinant of the variance-covariance matrix of an estimator of a vector of parametric functions is referred to as the generalized variance of the estimator.

2.8 Estimation of the Residual Variance

We now consider the estimation of σ^2 for Model II.

Definition. A square matrix A is said to be an idempotent matrix if $A^2 = A$.

Lemma 2.5. *For any square matrix A such that $A^2 = mA$ for some scalar m,*

$$\mathrm{tr}(A) = m\,\mathrm{rank}(A).$$

Corollary. If A is idempotent, then $\mathrm{tr}(A) = \mathrm{rank}(A)$.

Lemma 2.6. *For any matrix A, P_A and $I - P_A$ are both symmetric idempotent matrices, where $P_A = A(A'A)^{-}A'$.*

Lemma 2.7. *For any $n \times p$ matrix A, $\mathrm{rank}(P_A) = \mathrm{rank}(A)$ and $\mathrm{rank}(I - P_A) = n - \mathrm{rank}(A)$.*

Lemma 2.8. *If $\underline{x}$ is an $n \times 1$ random vector having mean vector $\underline{\mu}$ and variance-covariance matrix Σ and if A is an $n \times n$ matrix of constants, then*

$$E(\underline{x}'A\underline{x}) = \mathrm{tr}(A\Sigma) + \underline{\mu}'A\underline{\mu}.$$

Let $\widehat{\underline{\beta}}$ be any solution to the normal equations. Under Model II,

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*)$$

is an unbiased estimator of σ^2.

2.9 Constrained Linear Models

Consider the case where $\underline{\beta}$ is restricted to the subspace of R^p consisting of $\underline{\beta}$ values that satisfy

$$A\underline{\beta} = \underline{b},$$

where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and $\underline{b}$ is a given $q \times 1$ vector. To avoid trivialities, we assume that the linear system $A\underline{\beta} = \underline{b}$ is consistent. We will refer to those modified versions of Models I—VII in which $\underline{\beta}$ is restricted to this subspace as Models Ia—VIIa.

Definition. Under Models IIa—VIIa, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Lemma 2.9. *Under Models IIa—VIIa, a linear estimator $c+\underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that*

$$c = \underline{k}'\underline{b} \quad \text{and} \quad \underline{\lambda}' = \underline{a}'X + \underline{k}'A.$$

Definition. Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, if there exists a linear estimator $c + \underline{a}'\underline{y}$ such that $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Theorem 2.11. *Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}$.*

Definition. Under Models IIa—VIIa, a least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, is that the constrained normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

have a solution whose "first" component is $\widehat{\underline{\beta}}$.

Lemma 2.10. *The matrix equation $BW = D$ is consistent, that is, has a solution for W, if and only if $\mathcal{C}(D) \subset \mathcal{C}(B)$ if and only if $\mathcal{C}^{\perp}(B) \subset \mathcal{C}^{\perp}(D)$ that is, if and only if $\underline{a}'D = \underline{0}'$ for every vector $\underline{a}$ such that $\underline{a}'B = \underline{0}'$.*

Theorem 2.12. *The constrained normal equations are consistent.*

Theorem 2.13. *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.*

Lemma 2.11.

$$\text{rank}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} + \text{rank}(A).$$

Lemma 2.12. *(i) $\mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$ is the same as $\mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}$. (ii)* $\text{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} = \text{rank}\begin{bmatrix} X \\ A \end{bmatrix}$.

Corollary to Lemma 2.11.

$$\text{rank}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank}\begin{bmatrix} X \\ A \end{bmatrix} + \text{rank}(A).$$

Theorem 2.14. *Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$. Then,*

(i) $A'G_{12}'A' = A'G_{21}A' = A'$;

(ii) $-A'G_{22}A = -A'G_{22}'A = X'XG_{21}'A = A'G_{21}X'X = X'XG_{12}A = A'G_{12}'X'X$;

(iii) $AG_{11}A'$, $AG_{11}X'X$, and $X'XG_{11}A'$ are all zero matrices;

(iv) $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG_{11}'X'XG_{11}X'X = X'XG_{11}'X'X$;

(v) $\text{tr}(X'XG_{11}) = \text{rank}\begin{bmatrix} X'X \\ A \end{bmatrix} - \text{rank}(A) = \text{tr}(X'XG_{11}')$;

(vi) $X'XG_{11}X'X$ and $A'G_{22}A$ are invariant for any choice of G_{11} and G_{22}.

Lemma 2.13. *For any generalized inverse*

$$G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \quad \text{of} \quad \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix},$$

$X'X = X'XG_{11}X'X - A'G_{22}A$.

Theorem 2.15. *Under Models IIa—VIIa, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations.

Lemma 2.14. *Under Models IIa—VIIa, a linear function* $\underline{\lambda}'\underline{\beta}$ *is estimable if and only if* $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$.

The equations that play the role of the conjugate normal equations in our development are the constrained conjugate normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

Theorem 2.16. *Under Models IIa—VIIa,* $\underline{\lambda}'\underline{\beta}$ *is estimable, that is,*

$$\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}, \quad \textit{that is,} \quad \underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix},$$

if and only if the constrained conjugate normal equations are consistent.

Corollary. Under Models IIa—VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}^{-}\begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

Theorem 2.17. *Under Models IIa—VIIa, the least squares estimate of an estimable function* $\underline{\lambda}'\underline{\beta}$ *is given by*

$$\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b},$$

where $\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix}$ *is any solution to the constrained conjugate normal equations.*

Suppose that the two vectors $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable. Let $\widehat{\underline{\beta}}$ represent any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$. Then, under Model IIa,

$$\operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'G_{11}\Lambda_2,$$

where $\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ is any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.

Theorem 2.18. *Under Models IIa—VIIa, if* $\underline{\lambda}'\underline{\beta}$ *is estimable and* $\widehat{\underline{\beta}}$ *is the first component of any solution to the constrained normal equations, then* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is a linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$. *Under Model IIa, the variance of* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is uniformly less than that of any other linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$, *that is,* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is the best (minimum variance) linear unbiased estimator (BLUE) of* $\underline{\lambda}'\underline{\beta}$.

Take $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ to be any solution to the constrained normal equations. For the residual sum of squares, we have

$$Q(\widehat{\underline{\beta}}) = (\underline{y} - X\widehat{\underline{\beta}})'\underline{y} - \underline{\rho}'\underline{b}.$$

Under Model IIa,

$$\widehat{\sigma}^2 = (\underline{y}'\underline{y} - \widehat{\underline{\beta}}' X'\underline{y} - \underline{\rho}'\underline{b})/(n-f)$$

is an unbiased estimator of σ^2, where $f = \text{rank}\begin{bmatrix} X \\ A \end{bmatrix} - \text{rank}(A)$.

Sometimes, we might wish to impose q inequality constraints, say

$$A\underline{\beta} \geq \underline{b}.$$

Here, the least squares problem is that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ subject to the constraints $A\underline{\beta} \geq \underline{b}$ or, equivalently, that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ and $\underline{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_q \end{bmatrix}$ subject to the constraints

$$\begin{aligned} A\underline{\beta} &= \underline{b} + \underline{u}, \\ u_i &\geq 0, \quad i = 1, \ldots, q. \end{aligned}$$

The quantities $u_1, \ldots, u_q$ are called surplus variables. By taking advantage of our results on the least squares problem with equality constraints, this problem can be reduced essentially to that of minimizing the convex function

$$\left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]' \left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]$$

with respect to $\underline{u}$ subject to the linear equality constraints $(I - AA^-)(\underline{b} + \underline{u}) = \underline{0}$ and the inequality constraints $u_i \geq 0, i = 1, \ldots, q$. This is a quadratic programming problem.

2.10 Extensions to Model III

Lemma 2.15. *Corresponding to any $n \times n$ nonnegative definite matrix A of rank $r \geq 1$, there exists an $r \times n$ matrix S such that*

$$A = S'S.$$

Moreover, for any $r \times n$ matrix S such that $A = S'S$, $\text{rank}(S) = r$.

In Model III, $\text{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix. Assume that H is nonsingular, and take R to be an $n \times n$ matrix such that $H = R'R$. Then, R is nonsingular. Let $\underline{z} = (R^{-1})'\underline{y}$ and $W = (R^{-1})'X$. Then,

$$\underline{y} = X\underline{\beta} + \underline{\epsilon} \iff \underline{z} = W\underline{\beta} + (R^{-1})'\underline{\epsilon}.$$

We will refer to the model

$$\underline{z} = W\underline{\beta} + \underline{h}$$

as the "transformed model," where the distribution of $\underline{h}$ is assumed to be identical to that of $(R^{-1})'\underline{\epsilon}$ under (the original) Model III and where the parameter space is taken to be $\{\underline{\beta}, \sigma^2 : \underline{\beta} \in R^p, \sigma^2 > 0\}$. The transformed model is of the same general form as Model II, with $E(\underline{h}) = \underline{0}$ and $\text{Var}(\underline{h}) = \sigma^2 I$.

Take $\widehat{\underline{\beta}}$ to be any solution to the equations

$$W'W\widehat{\underline{\beta}} = W'\underline{z},$$

or equivalently to the equations

$$X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y},$$

which are known as the Aitken equations.

Theorem 2.19. *Under Model III, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the Aitken equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator.*

Note that $\widehat{\underline{\beta}}$ is a solution to the Aitken equations if and only if $(\underline{z} - W\underline{\beta})'(\underline{z} - W\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, or equivalently if and only if $(\underline{y} - X\underline{\beta})'H^{-1}(\underline{y} - X\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$. The problem of minimizing

$$(\underline{y} - X\underline{\beta})'A(\underline{y} - X\underline{\beta})$$

with respect to $\underline{\beta}$ is called generalized least squares, where A is a positive definite matrix.

If $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable vectors and if $\widehat{\underline{\beta}}$ is a solution to the Aitken equations, then

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'H^{-1}X)^-\Lambda_2.$$

Also, an unbiased estimator of σ^2 is

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'H^{-1}(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*).$$

We now consider the estimation of an estimable function $\underline{\lambda}'\underline{\beta}$ under Model III for the case where $H \neq 0$ is possibly singular.

Theorem 2.20. *Under Model III, a linear estimator* $c+\underline{a}'\underline{y}$ *of an estimable function* $\underline{\lambda}'\underline{\beta}$ *is a BLUE (that is, has minimum variance in the class of linear unbiased estimators) if and only if* $c = 0$ *and* $\underline{a}$ *is the first component of some solution to the equations*

$$\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix}.$$

Corollary 1. Under Model III, $c+\underline{a}'\underline{y}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if $H\underline{a} \in \mathcal{C}(X)$ and $E(c+\underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$.

Corollary 2. Under Model III, if $c + \underline{a}'\underline{y}$ and $k + \underline{b}'\underline{y}$ are both BLUEs of the same estimable function, then $\text{Var}[c + \underline{a}'\underline{y} - (k + \underline{b}'\underline{y})] = 0$ or equivalently $c + \underline{a}'\underline{y} = k + \underline{b}'\underline{y}$ with probability 1.

Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$. Let $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ be estimable vectors. Then, the elements of $\Lambda_1'G_{12}'\underline{y}$ and $\Lambda_2'G_{12}'\underline{y}$ are BLUEs of the elements of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, respectively, and

$$\text{cov}[\Lambda_1'G_{12}'\underline{y}, \Lambda_2'G_{12}'\underline{y}] = \sigma^2\Lambda_1'(-G_{22})\Lambda_2.$$

An unbiased estimator of σ^2 is given by

$$\widehat{\sigma}^2 = \underline{y}'G_{11}\underline{y}/\left[\text{rank}(H, X) - \text{rank}(X)\right].$$

Model IIIa, where q constraints $A\underline{\beta} = \underline{b}$ are imposed on the parameter space, can be approached as a special case of Model III. The vector $\underline{b}$ can be regarded as a vector of q additional "data points," in which case the vector of all data points is given by $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$.

We have

$$\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix} = \begin{bmatrix} X \\ A \end{bmatrix} \underline{\beta} + \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix}$$

and

$$E\begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \underline{0} \text{ and } \text{Var}\begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \sigma^2\begin{bmatrix} H & 0 \\ 0 & 0 \end{bmatrix}.$$

If we take the parameter space to be $\left\{\underline{\beta}, \sigma^2 : \underline{\beta} \in R^p, \sigma^2 > 0\right\}$, we arrive at an unconstrained model for $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$ having the form of Model III. It can be verified that BLUEs of any estimable function $\underline{\lambda}'\underline{\beta}$ computed on the basis of this model are BLUEs of $\underline{\lambda}'\underline{\beta}$ under the original, constrained model, and vice versa.

2.11 When Is the Simple Least Squares Estimator a Best Linear Unbiased Estimator?

Take the model to be Model III, and suppose that $\underline{\lambda}'\underline{\beta}$ is an estimable function.

Theorem 2.21. *Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that $HX\underline{r} = X\underline{q}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.*

Corollary 1. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a matrix Q such that $HX = XQ$.

Corollary 2. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $HP_X = P_X H$ (that is, HP_X is a symmetric matrix).

Remark. Theorem 2.21 and its two corollaries are still valid if $\mathrm{Var}(\underline{y}) = \sigma^2 H$ is put in place of H. Furthermore, these results are still valid under Model VII if V is put in place of H.

2.12 Best Linear Unbiased Prediction

Suppose that the model for the data vector $\underline{y}$ is Model III, with H nonsingular, and that we wish to predict (or estimate) $\underline{\lambda}'\underline{\beta} + u$, where u is a random variable having zero expectation and finite variance. For example, $\underline{\lambda}'\underline{\beta} + u$ might correspond to a future data point.

A linear predictor (or estimator) $c + \underline{a}'\underline{y}$ of $\underline{\lambda}'\underline{\beta} + u$ is said to be unbiased if

$$E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta} \quad \text{for all} \quad \underline{\beta},$$

that is, if $E(c+\underline{a}'\underline{y}) = E(\underline{\lambda}'\underline{\beta}+u)$ for all $\underline{\beta}$. If there exists a linear unbiased predictor (or estimator) of $\underline{\lambda}'\underline{\beta} + u$, then $\underline{\lambda}'\underline{\beta} + u$ is said to be predictable (or estimable).

Suppose that $\mathrm{cov}(\underline{\epsilon}, u) = \sigma^2\underline{k}$, for some given vector $\underline{k}$, and assume that $\underline{\lambda}'\underline{\beta} + u$ is predictable (or estimable). The mean squared error of any linear unbiased predictor of $\underline{\lambda}'\underline{\beta} + u$, $c + \underline{a}'\underline{y}$, is

$$E\left[c + \underline{a}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right]^2 = \mathrm{Var}\left[c + \underline{a}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right].$$

Among linear unbiased predictors of $\underline{\lambda}'\underline{\beta} + u$, the predictor $\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$ has uniformly smaller mean squared error than any other linear unbiased predictor, where $\widehat{\underline{\beta}}$ is any solution to $X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y}$ and where $\widehat{u} = \underline{k}'H^{-1}(\underline{y} - X\widehat{\underline{\beta}})$. That is,

$$\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u} = \underline{b}'\underline{y} \equiv \left[\underline{k}'H^{-1} + (\underline{\lambda}' - \underline{k}'H^{-1}X)(X'H^{-1}X)^{-}X'H^{-1}\right]\underline{y}$$

is the best linear unbiased predictor of $\underline{\lambda}'\underline{\beta} + u$.

3 Multivariate Normal Distributions; Noncentral χ^2 and F Distributions; Distributions of Quadratic Forms

3.1 Multivariate Normal Distributions

Definition. Let $\underline{x}$ represent an $n \times 1$ random vector. The moment generating function (mgf) of $\underline{x}$, denoted by $m_{\underline{x}}(\cdot)$, is defined by

$$M_{\underline{x}}(\underline{t}) = E\left[\exp(\underline{t}'\underline{x})\right]$$

if the expectation exists for $-h < t_i < h$ $(i = 1, \ldots, n)$ for some number $h > 0$, where $\underline{t} = (t_1, \ldots, t_n)'$. If this expectation does not exist, then $\underline{x}$ has no mgf.

Theorem 3.1. *Consider two $n \times 1$ random vectors $\underline{x}_1$ and $\underline{x}_2$, and assume that the mgf of each exists. The cumulative distribution functions (cdf's) of $\underline{x}_1$ and $\underline{x}_2$ are identical if and only if the mgf's are equal for all values of $\underline{t}$ in an open rectangle that includes the origin.*

Theorem 3.2. *If random vectors $\underline{x}, \underline{x}_1, \ldots, \underline{x}_m$, where $\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}$, have mgf's denoted by $M_{\underline{x}}(\cdot)$, $M_{\underline{x}_1}(\cdot), \ldots, M_{\underline{x}_m}(\cdot)$, then $\underline{x}_1, \ldots, \underline{x}_m$ are mutually independent if and only if the mgf's satisfy*

$$M_{\underline{x}}(\underline{t}) = M_{\underline{x}_1}(\underline{t}_1) M_{\underline{x}_2}(\underline{t}_2) \cdots M_{\underline{x}_m}(\underline{t}_m) = \prod_{i=1}^{m} M_{\underline{x}_i}(\underline{t}_i)$$

for all values of $\underline{t} = \begin{bmatrix} \underline{t}_1 \\ \vdots \\ \underline{t}_m \end{bmatrix}$ in an open rectangle that includes the origin.

Theorem 3.3. *Let A and B be $n \times n$ matrices of constants, $\underline{a}$ and $\underline{b}$ be $n \times 1$ vectors of constants, a_0 and b_0 be constants, and $\underline{x} = (x_1, \ldots, x_n)'$ be a vector of variables. If A is symmetric and B is positive definite, then*

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} (\underline{x}' A \underline{x} + \underline{x}' \underline{a} + a_0) \exp\left[-(\underline{x}' B \underline{x} + \underline{x}' \underline{b} + b_0)\right] dx_1 \cdots dx_n$$

$$= \left(\frac{1}{2}\right) \pi^{n/2} |B|^{-1/2} \left[\operatorname{tr}(AB^{-1}) - \underline{b}' B^{-1} \underline{a} + \left(\frac{1}{2}\right) \underline{b}' B^{-1} A B^{-1} \underline{b} + 2a_0\right] \exp\left[\left(\frac{1}{4}\right) \underline{b}' B^{-1} \underline{b} - b_0\right].$$

Definition. A random variable z is said to have a univariate standard normal distribution if the probability density function (pdf) of z is given by

$$n(z\colon 0, 1) = (2\pi)^{-1/2} \exp\left(-z^2/2\right), \quad z \in R,$$

or, equivalently, if the mgf of z is given by $M_z(t) = \exp(t^2/2)$, $t \in R$

Definition. A $k \times 1$ random vector $\underline{z}$ is said to have a $\underline{k\text{-variate standard normal distribution}}$ if the pdf of $\underline{z}$ is given by

$$n(\underline{z}\colon \underline{0}, I) = (2\pi)^{-k/2} \exp\left[-\left(\frac{1}{2}\right)\underline{z}'\underline{z}\right], \quad \underline{z} \in R^k.$$

Definition. Let $\underline{x}$ be a $p \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ, where $\text{rank}(\Sigma) = k \geq 1$. Let Γ be a $k \times p$ matrix such that $\Sigma = \Gamma'\Gamma$. The random vector $\underline{x}$ is said to have a $\underline{p\text{-variate normal distribution of rank } k}$ if the cdf of $\underline{x}$ is the same as that of the random vector $\underline{\mu} + \Gamma'\underline{z}$, where $\underline{z}$ is a $k \times 1$ random vector with the pdf $n(\underline{z}\colon \underline{0}, I)$.

Notation. We use $N(\underline{x}\colon \underline{\mu}, \Sigma)$ to represent the cdf of a random vector $\underline{x}$ having a (multivariate) normal distribution with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. We use $\underline{x} \sim N(\underline{\mu}, \Sigma)$ to indicate that $\underline{x}$ is a random vector whose cdf is $N(\underline{x}\colon \underline{\mu}, \Sigma)$.

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. The mgf of $\underline{x}$ is given by

$$M_{\underline{x}}(\underline{t}) = \exp\left[\underline{t}'\underline{\mu} + \left(\frac{1}{2}\right)\underline{t}'\Sigma\underline{t}\right], \quad \underline{t} \in R^p.$$

Theorem 3.4. *Let $\underline{x} = (x_1, \ldots, x_p)' \sim N(\underline{\mu}, \Sigma)$, where $\text{rank}(\Sigma) = p$. Then, $\underline{x}$ has a pdf and it is given by*

$$n(\underline{x}\colon \underline{\mu}, \Sigma) = (2\pi)^{-p/2}|\Sigma|^{-1/2} \exp\left[-\left(\frac{1}{2}\right)(\underline{x} - \underline{\mu})'\Sigma^{-1}(\underline{x} - \underline{\mu})\right], \quad \underline{x} \in R^p.$$

Theorem 3.5. *Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Let B be a $q \times p$ matrix of constants and $\underline{b}$ be a $q \times 1$ vector of constants. Then, $\underline{y} = B\underline{x} + \underline{b} \sim N(B\underline{\mu} + \underline{b}, B\Sigma B')$.*

Theorem 3.6. *Suppose that $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Partition $\underline{x}, \underline{\mu}$, and Σ conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. Then, $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$ for $i = 1, \ldots, m$.

Theorem 3.7. *Suppose that $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Partition $\underline{x}, \underline{\mu}$, and Σ conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. Then, $\underline{x}_1, \ldots, \underline{x}_m$ are mutually independent if and only if $\Sigma_{ij} = 0$ for $i = 1, \ldots, m$, $j = 1, \ldots, m$, and $i \neq j$.

Corollary. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Let B_1 and B_2 be $q \times p$ and $r \times p$ matrices of constants. Let $\underline{b}_1$ and $\underline{b}_2$ be $q \times 1$ and $r \times 1$ vectors of constants. Then, the random vectors

$$\underline{y}_1 = B_1\underline{x} + \underline{b}_1 \quad \text{and} \quad \underline{y}_2 = B_2\underline{x} + \underline{b}_2$$

are independent if and only if $B_1\Sigma B_2^{'} = 0$.

Theorem 3.8. *Let* $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ *be a partitioned matrix such that* M*,* A*, and* B *are square matrices. If* B *is nonsingular, then* $|M| = |B| \cdot |A - CB^{-1}D|$*. Similarly, if* A *is nonsingular, then* $|M| = |A| \cdot |B - DA^{-1}C|$*.*

Theorem 3.9. *Suppose that* $\underline{x} \sim N(\underline{\mu}, \Sigma)$ *and* Σ *is nonsingular. Partition* $\underline{x}, \underline{\mu}$*, and* Σ *conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \underline{x}_2 \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}.$$

Then, the conditional distribution of $\underline{x}_1$ *given* $\underline{x}_2$ *is normal with mean vector* $\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2)$ *and variance-covariance matrix* $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$*.*

3.2 The Moment Generating Function and Variance of a Quadratic Form

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Take A to be a $p \times p$ symmetric matrix of constants. Then, for t such that $|t|$ is sufficiently small,

$$M_{\underline{x}^{'}A\underline{x}}(t) = |I - 2tA\Sigma|^{-1/2} \exp\left[t\underline{\mu}^{'}\left(I - 2tA\Sigma\right)^{-1} A\underline{\mu}\right].$$

The variance of $\underline{x}^{'}A\underline{x}$ is given by

$$\text{Var}(\underline{x}^{'}A\underline{x}) = 2\text{tr}\left[(A\Sigma)^2\right] + 4\underline{\mu}^{'}A\Sigma A\underline{\mu}.$$

Lemma 3.1. *If* A *is a* $p \times p$ *positive definite matrix and* B *is a* $p \times p$ *symmetric matrix, then there exists a* $\delta > 0$ *such that, for* $|t| < \delta$*,* $A + tB$ *is positive definite.*

Theorem 3.10. *Let* R *be a* $p \times p$ *matrix,* T *a* $q \times q$ *matrix,* S *a* $p \times q$ *matrix, and* U *a* $q \times p$ *matrix. If* R *and* T *are nonsingular, then*

$$|R + STU| = |R| \cdot |T| \cdot |T^{-1} + UR^{-1}S|.$$

Corollary. Let S be a $p \times q$ matrix and U a $q \times p$ matrix. Then, $|I + SU| = |I + US|$.

3.3 The Noncentral χ^2 and F Distributions

Definition. A random variable u is said to have a noncentral χ^2 distribution with degrees of freedom n and noncentrality parameter λ if u has the pdf

$$\chi^2(u\colon n,\lambda)=\begin{cases}\displaystyle\sum_{j=0}^{\infty}\left(\frac{e^{-\lambda}\lambda^j}{j!}\right)\left[\frac{u^{(n+2j-2)/2}e^{-u/2}}{\Gamma\left(\frac{n+2j}{2}\right)2^{j+(n/2)}}\right], & \text{for } u>0,\\ \\ 0, & \text{for } u\le 0,\end{cases}$$

where $\lambda\ge 0$ and $n>0$.

Note. In the above pdf, $0!=1$ and $0^0=1$.

With $\lambda=0$, the noncentral χ^2 distribution reduces to the χ^2 or central χ^2 distribution with the pdf

$$\chi^2(u\colon n)=\begin{cases}\frac{u^{(n-2)/2}e^{-u/2}}{\Gamma\left(\frac{n}{2}\right)2^{n/2}}, & \text{for } u>0,\\ \\ 0 & \text{for } u\le 0.\end{cases}$$

Notation. We use $u\sim\chi^2(n,\lambda)$ to indicate that u is a random variable whose pdf is $\chi^2(u\colon n,\lambda)$, and $u\sim\chi^2(n)$ will be used if the pdf is $\chi^2(u\colon n)$.

Let $u\sim\chi^2(n,\lambda)$. We have

$$M_u(t)=(1-2t)^{-n/2}\exp\left(\frac{2t\lambda}{1-2t}\right),\quad t<\frac{1}{2}.$$

Theorem 3.11. *Let $\underline{x}\sim N(\underline{\mu},I_n)$. Define w by $w=\underline{x}'\underline{x}$. Then, $w\sim\chi^2(n,\frac{1}{2}\underline{\mu}'\underline{\mu})$.*

Let $u\sim\chi^2(n,\lambda)$. Then,

$$E(u)=n+2\lambda \quad\text{and}\quad \mathrm{Var}(u)=2(n+4\lambda).$$

Theorem 3.12. *If $u_1,\ldots,u_m$ are mutually independent random variables and if $u_i\sim\chi^2(n_i,\lambda_i)$ for $i=1,\ldots,m$, then $u=u_1+\cdots+u_m\sim\chi^2(n,\lambda)$, where $n=n_1+\cdots+n_m$ and $\lambda=\lambda_1+\cdots+\lambda_m$.*

Theorem 3.13. *Let $\underline{x}\sim N(\underline{\mu},\Sigma)$, where $\underline{x}$ is an $n\times 1$ vector and $\mathrm{rank}(\Sigma)=n$. Define w by $w=\underline{x}'\Sigma^{-1}\underline{x}$. Then, $w\sim\chi^2(n,\frac{1}{2}\underline{\mu}'\Sigma^{-1}\underline{\mu})$.*

Definition. Let u_1 and u_2 be independent random variables such that $u_1\sim\chi^2(n_1,\lambda)$ and $u_2\sim\chi^2(n_2)$, where $\lambda\ge 0$, $n_1>0$, and $n_2>0$. Let

$$w=\frac{u_1/n_1}{u_2/n_2}.$$

The distribution of w is called a noncentral F distribution with degrees of freedom n_1 and n_2 and noncentrality parameter λ. With $\lambda = 0$, the noncentral F distribution reduces to the F or central F distribution.

An mgf does not exist for the noncentral F distribution (including the central F distribution).

3.4 Distributions of Quadratic Forms

In our discussion of distributions of quadratic forms, we assume that the matrix A associated with any quadratic form $\underline{x}'A\underline{x}$ is symmetric.

Theorem 3.14. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for a quadratic form $\underline{x}'A\underline{x}$ and a vector $B\underline{x}$ of linear forms to be independent is that $B\Sigma A = 0$.*

Theorem 3.15. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A_1\underline{x}$, ..., $\underline{x}'A_k\underline{x}$ to be mutually independent is that $A_i\Sigma A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$.*

Corollary. If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A\underline{x}$ and $\underline{x}'B\underline{x}$ to be independent is that $B\Sigma A = 0$.

Remarks. If Σ is nonsingular, the condition $B\Sigma A = 0$ in Theorem 3.14 and the corollary to Theorem 3.15 is necessary as well as sufficient. If Σ is singular, the condition is no longer necessary.